

CSCE 650 VIRTUAL REALITY · MODULE 02 · LECTURE NOTES

Human Vision for VR

The specification you are actually building against.

MODULE OUTCOMES

- State the relevant limits of human vision in numbers.
- Explain acuity falloff and what it permits.
- Distinguish the depth cues and say which VR provides.
- Explain the vergence-accommodation conflict.
- Explain flicker fusion and persistence.

1 The numbers

Property	Human visual system	Typical consumer headset
Horizontal field of view	About 200° total, of which roughly 120° is binocular.	100–110° , with the periphery absent entirely.
Foveal acuity	About 60 pixels per degree for someone with 20/20 vision — the limit at which a pixel grid becomes invisible.	15–35 pixels per degree. The gap is the reason the screen door effect and visible aliasing persist.
Acuity 20° off-axis	Roughly a sixth of foveal.	Identical to the centre, since rendering is uniform — which is pure waste.
Flicker fusion threshold	60–90 Hz, and higher in the periphery than at the centre.	72–120 Hz refresh.
Useful stereo depth range	To about 10 m; beyond that the disparity is too small to use.	Same — and beyond 10 m you are relying on monocular cues, which is worth knowing when designing distant content.

'**Retinal resolution**' — the point at which the display ceases to be the limiting factor — means roughly 60 pixels per degree across the full 200°. That is on the order of ten times the pixel count of current headsets *per eye*, which is a useful thing to know when evaluating claims about how close the technology is.

ACUITY COLLAPSES OUTSIDE THE FOVEA

The fovea is about 2° across — approximately a thumbnail held at arm's length — and essentially all detailed vision happens there. It is also where the cone density is highest by a large factor.

At 20° off-axis acuity has fallen to roughly a sixth of foveal, and it continues to drop steeply. At the edge of the field it is negligible.

A headset nonetheless renders the entire field at uniform resolution. The overwhelming majority of the pixels it produces are far more detailed than the eye looking at them can resolve — the work is done and then discarded by the visual system.

Foveated rendering (Module 10) recovers that waste, either fixed (assuming the user looks mostly forward) or eye-tracked. The entire justification for the technique is on this page.

THE PERIPHERY IS POOR AT DETAIL AND EXCELLENT AT MOTION

A demonstration worth doing: hold a playing card at arm's length, fix your gaze straight ahead, and move the card sideways. You will be unable to identify the card past roughly 10° — and yet the world does not appear blurry at the edges.

The brain fills in. The vivid, uniformly detailed visual field you experience is substantially constructed from memory, from rapid saccades you are unaware of, and from assumption.

But peripheral sensitivity to motion and flicker is higher than foveal, not lower. This is evolutionarily sensible — the periphery is a threat detector — and it has direct design consequences.

So detail can be reduced in the periphery; motion and flicker cannot. That asymmetry governs both foveated rendering (Module 10) and comfort design (Module 07), where peripheral motion is precisely what induces vection and therefore sickness.

2 Depth

Cue	Mechanism	Useful range	Provided by VR?
Stereopsis	Disparity between the two retinal images.	To about 10 m.	Yes — the medium's headline feature.
Motion parallax	Nearer objects move further across the retina as the head moves.	All distances.	Yes, and underrated. See below.
Vergence	The inward rotation of the eyes.	To about 2 m.	Yes, and in conflict with accommodation.
Accommodation	The focusing effort of the lens.	To about 2 m.	No. The display sits at one fixed optical distance.
Occlusion	Nearer objects hide further ones.	All.	Yes — and it is the strongest cue of all, which is why penetration breaks presence so completely (Module 01).

Cue	Mechanism	Useful range	Provided by VR?
Relative size, linear perspective, texture gradient	Geometric.	All.	Yes, for free.
Shading and shadows	Light falloff and contact shadows.	All.	Yes — and the cheapest depth cue you have. A contact shadow under an object does more for its apparent position than a great deal of geometric accuracy.

Stereopsis receives most of the attention and motion parallax does more of the work. A user moving their head even slightly gets continuous, high-precision depth information across the entire range, where stereo is useful only to about ten metres. This is why a six-degree-of-freedom headset feels so much more convincing than a three-degree-of-freedom one, even though both provide stereo.

THE VERGENCE-ACCOMMODATION CONFLICT

Vergence is the inward rotation of the eyes to converge on an object. **Accommodation** is the change in the lens's shape to bring that distance into focus.

In the real world the two are coupled — neurally, as well as physically. The brain uses each as a cue to drive the other, and they are always consistent.

In a headset the display is at a single fixed optical distance, typically 1.5 to 2 m. So when a virtual object appears at 30 cm, the eyes verge as though it were at 30 cm while continuing to accommodate at 2 m. The two signals disagree, persistently.

The result is eye strain and fatigue, and it is the principal reason that sustained close-up work in VR is uncomfortable. **No shipping consumer headset solves it.** Varifocal displays (which move the focal plane to match vergence) and light-field displays (which present a genuine focal range) are the active research directions, and neither has reached production.

- **Keep interactive content between about 0.5 and 2 m**, near the display's focal distance. **This is the single most effective mitigation** and it explains a great deal of VR interface design — why menus hover at arm's length rather than close to the face.
- **Avoid sustained close work.** Reading text at 30 cm becomes uncomfortable within minutes, however crisp the text is.
- **Place UI at a consistent, comfortable depth.** A UI element at a different depth from the object it labels forces repeated vergence changes, which is tiring in a way users cannot articulate.
- **Consider reducing stereo separation for very close objects.** It is physically incorrect and measurably more comfortable, and several shipped applications do it.
- **Place nothing closer than roughly 20 cm.** Below that, binocular fusion fails outright and the image doubles.

3 Time, flicker, and persistence

LOW PERSISTENCE IS WHY MODERN VR IS NOT A SMEARED MESS

A full-persistence display holds each frame illuminated for the entire frame period. When the head turns, the eye tracks the world and therefore sweeps across a stationary image during those 11 milliseconds — and the result is smeared across the retina. Early headsets had this problem badly, and it was one of the main reasons they felt wrong.

A low-persistence display illuminates for roughly 2 ms per frame and is dark for the remainder. Each frame is a brief flash at a definite position, so the eye receives a sharp image regardless of head motion.

Eye-motion smear disappears. This was one of the decisive changes between the first-generation and modern headsets, and it is why the image stays sharp when you turn your head quickly.

The costs are brightness and flicker. A display lit for 2 ms in every 11 is, other things equal, considerably dimmer — and the flicker becomes perceptible, particularly in the periphery where flicker sensitivity is highest (§1). **This is a significant part of why 90 Hz rather than 72 Hz is the comfortable target.**

- **Do not add motion blur.** The display's low persistence exists specifically to eliminate motion blur; reintroducing it in a post-process is perverse, and it adds perceived latency because the blur trails behind the true position.
- **Be careful with temporal antialiasing.** TAA accumulates across frames and therefore smears during head motion — which in VR is constant rather than occasional. The smear reads to the user as latency even when the frame timing is correct.
- **MSAA is frequently the right choice in VR,** despite having fallen out of fashion in flat real-time rendering, precisely because it is a within-frame technique that accumulates nothing. **This reverses a decade of practice, and the reason is persistence.**
- **Avoid high-frequency flickering content,** and especially avoid it near the edges of the field where sensitivity is highest. Thin high-contrast geometry that aliases temporally is the usual culprit.
- **High-contrast edges and textures in the periphery induce vection** — the illusion of self-motion — and therefore sickness (Module 07). This is why comfort vignettes darken the periphery during movement.

BINOCULAR RIVALRY: THE TWO EYES MUST AGREE

If the two eyes receive sufficiently different images, the brain does not average them. **It alternates** — perceiving one, then the other, in an unstable rhythm. The experience is disorienting and quickly unpleasant.

The usual causes are bugs: an effect rendered into one eye's buffer only; a view-dependent effect such as a reflection computed from a single eye's position and applied to both; a UI element composited with incorrect or zero disparity; a post-process applied asymmetrically.

Note that some difference is correct. Specular highlights and reflections genuinely differ between the eyes, and that difference is part of what makes a surface look wet or polished. The problem is gross disagreement, not disagreement as such.

The test is trivial and worth doing routinely: close one eye, then the other, and compare. **If the two images differ in anything beyond viewpoint, something is wrong** — and this is a class of bug that is invisible with both eyes open, because the brain works hard to conceal it.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
LaValle — Virtual Reality, Chapters 4 and 6 (free)	Light, optics, and the visual physiology of this module, with the numbers and their sources.
Stanford EE267 — human visual system and stereo lectures (free)	The acuity, depth cue, and vergence-accommodation material, with demonstrations you can run.
Hoffman et al. — Vergence-accommodation conflicts hinder visual performance and cause visual fatigue (free)	The study that quantified §2's conflict. The measurements are what justify the 0.5–2 m rule.
Abrash — Down the VR rabbit hole: fixing judder (free)	Persistence, judder, and why low persistence mattered so much. The best explanation of §3 anywhere.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Measure your own visual acuity falloff: fix your gaze, have someone move a card of text sideways, and record the angle at which you can no longer read it.

- 2 Compute the pixels per degree of a headset you have access to, from its resolution and field of view. Compare against 60 ppd.
- 3 Compute how many pixels per eye a retinal-resolution headset would need at 200°, and what that implies for rendering cost.
- 4 In a VR scene, disable stereo (render both eyes from the same point). Describe what is lost, and then move your head and describe what is retained.
- 5 Add a contact shadow under a floating object and compare the apparent depth with and without. This is the cheapest depth cue and the difference is larger than expected.
- 6 Place text at 30 cm, 1 m, and 2 m in VR and read each for two minutes. Record the fatigue.
- 7 Find the distance at which binocular fusion fails for you, by moving a virtual object steadily toward your face.
- 8 Compare MSAA and TAA in a VR scene while turning your head. Describe the difference during motion rather than when still.
- 9 Deliberately render an effect into one eye only and experience binocular rivalry. Then fix it.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Give foveal acuity, human field of view, and typical headset figures for each.
- 2 How does acuity vary with eccentricity, and what technique does that enable?
- 3 What is the periphery good at and bad at, and what two design rules follow?
- 4 Name seven depth cues and say which VR provides.
- 5 Why does motion parallax do more work than stereopsis?
- 6 Explain the vergence-accommodation conflict and give four mitigations.
- 7 What is low persistence, what does it fix, and what does it cost?
- 8 Why is motion blur wrong in VR, and why might MSAA beat TAA?
- 9 What is binocular rivalry, what causes it, and how do you test for it?